

CHAEYUN KIM

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RESEARCH INTERESTS

Vision-Language Compositional Understanding, Robustness and Reasoning of (M)LLMs, AI Safety & Alignment, Red Teaming & Adversarial Attacks, Evaluation and Benchmarks

EDUCATION

Seoul National University (SNU)

Seoul, Korea

M.S. in Data Science

Mar. 2023 – Feb. 2025

- Relevant Coursework: Machine Learning & Deep Learning I, II, Big Data & Knowledge Management Systems I, II, Computing for Data Science I, Scalable High-Performance Computing, Data Science Project, Machine Learning for Visual Understanding, Conversational AI & Dialogue Systems

B.S. in International Relations

Mar. 2017 – Feb. 2022

- CGPA: 4.15/4.3, Graduated *Summa Cum Laude*
- Dean's List, Academic Scholarship for 5 semesters (2018-2020)

PUBLICATIONS

[Journal/Conference (including in-review)]

- Seongsu Ha*, **Chaeyun Kim***, Donghwa Kim*, Junho Lee, Sangho Lee, Joonseok Lee**, "Finding NeMo: Negative-mined Mosaic Augmentation for Referring Image Segmentation," European Conference on Computer Vision (ECCV). Springer, 2025. Available: <https://arxiv.org/pdf/2411.01494>
- **Chaeyun Kim***, Seunghoon Yi*, Yejin Kim, Yohan Jo, Joonseok Lee, "Towards Motion-aware Referring Image Segmentation". (*AISTATS 26, under review*)
- Joonki Min*, **Chaeyun Kim***, Hyungwook Choi, Yejin Kim, Kihyun Kim, Yohan Jo, Joonseok Lee, "Towards Robust Benchmark of Object Hallucination on Multiple Images". (*ICLR 26, under review*)
- **Chaeyun Kim**, Yongtaek Lim, Kihyun Kim, Junghwan Kim, Minwoo Kim, "CAGE: A Framework for Culturally Adaptive Red-Teaming Benchmark Generation". (*ICLR 26, under review*)
- Min-jae Jung, Yongtaek Lim, **Chaeyun Kim**, Junghwan Kim, Kihyun Kim, Minwoo Kim, "A Strategy-Response Multiplex Network Approach to Automated LLM Red Teaming" (*ARR 25 Oct. under review*)
- **Chaeyun Kim***, Daeyoung Park*, Junghwan Kim, Jinyoung Jeong, Eunji Song, Minwoo Kim, "FinRED: An Expert-Guided Red-Teaming Benchmark for Financial LLM Safety" (*WWW 26, under review*)

[Books]

- Kim, S.B., **Kim, C.Y.**, et al, "The 4th Industrial Revolution and U.S.-China Power Politics: A Perspective from Information Politics," Social Criticism Academy, May 2020 (authored subsection of Chp 8), [Link](#)
- Kim, U.Y., **Kim, C.Y.**, et al, "2020 Civic Politics Yearbook: Theories and Practices of Community-based Education," Pureungil, Jan. 2021 (authored Chapter 3), [Link](#)

RESEARCH EXPERIENCE

SelectStar (DATUMO Inc.)

Seoul, Korea

Red Teaming AI Researcher

May 2025 – Present

[1] "FinRED: Expert-Guided Red-Teaming Benchmark for Financial LLM Safety" (*WWW 2026, In review*)

- Developed the first comprehensive **financial domain red-teaming benchmark** in collaboration with **FSI (Financial Security Institute)**, featuring 2-level taxonomy of financial risks

- Created expert-validated finance-specific safety rubric demonstrating substantially higher alignment with human judgments compared to generic rubrics.

[2] “STAR-Teaming: Strategy-Response Multiplex Network for Automated LLM Red Teaming” (ARR Oct. Submission)

- Participated in automated LLM red teaming research (STAR-Teaming) by contributing to main experiments and paper writing.
- Responsible for refactoring and optimizing the research-level code for commercial product integration, bridging the gap between research and deployment.

[3] “CAGE : A Culturally Adaptive Red-Teaming Benchmark for LLMs” (ICLR 2026, in review)

- Pioneered the **CAGE framework**, which uses flexible ‘**Semantic Molds**’ as content guidelines to generate diverse and realistic adversarial prompts tailored to any cultural and legal context.
- Built **KorSET**, the first large-scale Korean safety benchmark, and demonstrated the framework's high **efficiency and scalability** (e.g., generating over 1,250 prompts in a single run).
- Validated the framework’s generalizability by extending it to low-resource languages (e.g. Khmer), quantitatively proving its superior efficacy over translation-based methods.

SNU Visual Information Processing Lab

Seoul, Korea

Graduate Researcher

Sep. 2024 – Feb. 2025

[1] “Towards Robust Benchmark of Object Hallucination on Multiple Images” (ICLR 2026, in review)

- Designed and is developing the **Multi-Image Object Hallucination (MIOH)** benchmark, featuring complex reasoning tasks (e.g., cross-image counting, comparative attribute/position queries).
- **Formulated a framework of five controllable adversarial pressures** (e.g., visual similarity, language prior, weak visual signals) to systematically probe model vulnerabilities.

[2] “Towards Motion-aware Referring Image Segmentation” (AISTATS 2026, in review)

- **Exposed a critical failure in state-of-the-art RIS models:** their significant underperformance on motion-centric queries due to an over-reliance on static visual cues.
- Proposed a unified framework that combines **text augmentation strategy** to supplement under-represented motion-centric phrases with novel **Multimodal Radial Contrastive Loss (MRaCL)** for context-aware semantic alignment.
- Introduced **M-Bench**, a novel benchmark for evaluating motion comprehension.

Graduate Research Assistant (Advisor: Prof. Joonseok Lee)

Jan. 2023 – Aug. 2024

- Designed a video frame interpolation model incorporating Fourier Transform loss to preserve high-frequency details in low-FPS sampled frames, using multi-resolution inter-frame deformable attention to enhance diverse object feature representation.
- Developed negative-mined mosaic augmentation for referring image segmentation, improving model comprehension of visual-linguistic cues in ambiguous scenarios; **co-first author (ECCV 2024)**.

PROJECT EXPERIENCE

Selectstar (DATUMO Inc.)

Seoul, Korea

“Reasoning-based LLM Safety Guardrail Development & Commercialization”

Lead AI Safety Researcher

Aug. 2025 – Present

- Led development of Korean reasoning-based safety guardrail overcoming limitations of rule-based judges (poor explainability, performance degradation on policy changes, high FP rates).
- Designed multi-task GRPO training pipeline using 4,500 carefully curated examples, achieving **91% accuracy** on internal safety benchmarks, outperforming SFT baselines.

- Successfully commercialized solution with **Woori Bank**; currently advancing with adaptive reasoning mechanisms for paper submission.

“Collaboration with Samsung SDS for AI Guardrail Safety Enhancement”

Lead AI Safety Researcher & Engineer

Sep. 2025 – Nov. 2025

- Led construction of 10,000 adversarial query-response pairs to strengthen Korean safety judgment capabilities of SDS guardrail system.
- Applied CAGE framework for generating base harmful queries, combined with STAR-Teaming and diverse jailbreaking techniques to create comprehensive attack scenarios.

"WBL AI Foundation Model Safety Consortium (SKT)"

Tech Lead & Engineer

Sep. 2025 – Dec. 2025

- Led construction of safety/red-teaming dataset in SKT's AI foundation model development project.
- Built multi-turn focused harmless datasets and RL-ready triplet format (harmful query, safe response, unsafe response) for robust post-training alignment.

Seoul National University - Graduate School of Data Science

Seoul, Korea

“ECOLENS: Real-time Object Recognition for Recycling Guidance”

Machine Learning Engineer & Back-end Developer

Sep. 2023 – Dec. 2023

- Developed a system providing user-customized text and speech recycling instructions for objects based on photos and information about available bins.
- Constructed the application backend using the Flask framework; implemented a MongoDB database for efficient data management.

“Data-Effective Semantic Segmentation on 4D Point Cloud Videos”

Team Member

Mar. 2023 – Jun. 2023

- Designed a data-loading pipeline optimized for processing large-scale datasets.
- Proposed a framework combining active learning and contrastive learning to efficiently utilize feature representations in irregular and unordered point clouds.

“GSDS PleaseGraduate: Graduation Requirement Checker and Course Recommender”

Team Leader

Oct. 2022 – Jan. 2023

- Spearheaded the development of a customized graduation requirement checking system for the Graduate School of Data Science.
- Designed user interface and led front-end development using HTML, CSS, jQuery, and Django; managed the data preprocessing pipeline.

EXCHANGE PROGRAMS

University of Washington, Seattle

Seattle, WA

Henry M. Jackson School of International Studies

Mar. 2020 – Jun. 2020

Vrije Universiteit Brussel

Brussels, Belgium

Institute for European Studies

Jul. 2019 – Aug. 2019

- Participated in SNU in the EU program; took a lecture on the European Union’s international mission.

TEACHING ASSISTANTSHIPS

“Core Computing: Thinking with Computers – Data Structure and Algorithms,” SNU (TA)

Fall. 2024

“Principles and Applications of Data Science,” SNU (TA)

Fall. 2023

“Machine Learning and Deep Learning I,” SNU (TA)	Fall. 2023
“ExploreCSR Program (with Google),” SNU (tutor, mentor)	Summer. 2023
“Basic Computing: First Adventures in Computing,” SNU (tutor)	Fall. 2022

SCHOLARSHIPS & AWARDS

Graduate

<i>Academic Scholarship : Full Tuition Exemption</i> , Seoul National University	Sep.2023
<i>Data Science Graduate School Scholarship</i> , Seoul National University	May.2023
4 th Stage BK21 Scholarship, Seoul National University	Mar.2023

Undergraduate

<i>International Scholarship</i> , SNU Office of International Affairs	Sep.2020
<i>Dean's List, Academic Scholarship</i> , Seoul National University	Sep.2018 – Dec.2020
<i>Global Leadership Award</i> , Seoul National University	Feb.2022
1 st Place, Division of Policy Strategy, Future Aerospace Conference	Oct.2021
1 st Place, Undergraduate Social Science Research Grant, SNU	Feb.2020
3 rd Place, KOICA 16th International Development Cooperation Paper Competition	May.2019

LEADERSHIP

<i>Vice President</i> , SNU Graduate School of Data Science Student Association	Aug. 2023 – Feb. 2024
- Served as the cohort's deputy leader and an active member, representing the Data Science department. - Organized various on-campus events, bi-weekly student seminars, and community-building activities in collaboration with school administrators.	

ACADEMIC ACTIVITIES

SNU-HIT International Joint Seminar	Sep. 2017 – Oct. 2017, Sep. 2019 – Oct. 2019
- Participated in the international exchange seminar between Seoul National University Department of Political Science & International Relations and Hitotsubashi University Faculty of Law 2017: Presented and led discussions on the “Korea-Japan Security Cooperation Plan” and “General Security of Military Information Agreement (GSOMIA) and Japan’s Constitutional Amendment.” 2019: Hosted seminars in the History Department; facilitated discussions on the Supreme Court’s ruling on forced labor, individual claim issues, and strategies to improve the Korea-Japan relations.”	
12th Asian Future Political Leaders’ Association (AFPLA)	May 2018 – Aug. 2018
Served as Politics Division Session Leader; presented and led discussions on “Definition, Importance, and the Evolution of 'Citizen' and 'Citizen Culture' in Past, Present, and Future Contexts.”	

PROFICIENCY IN SKILLS

Programming: Python, R, SQL, C/C++

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Sckit-learn, Seaborn, Flask

Languages: Korean (native), English (full-professional fluency), Mandarin Chinese, Spanish (intermediate)